Computational Insights into Modular Nucleobase GFP-Surrogates

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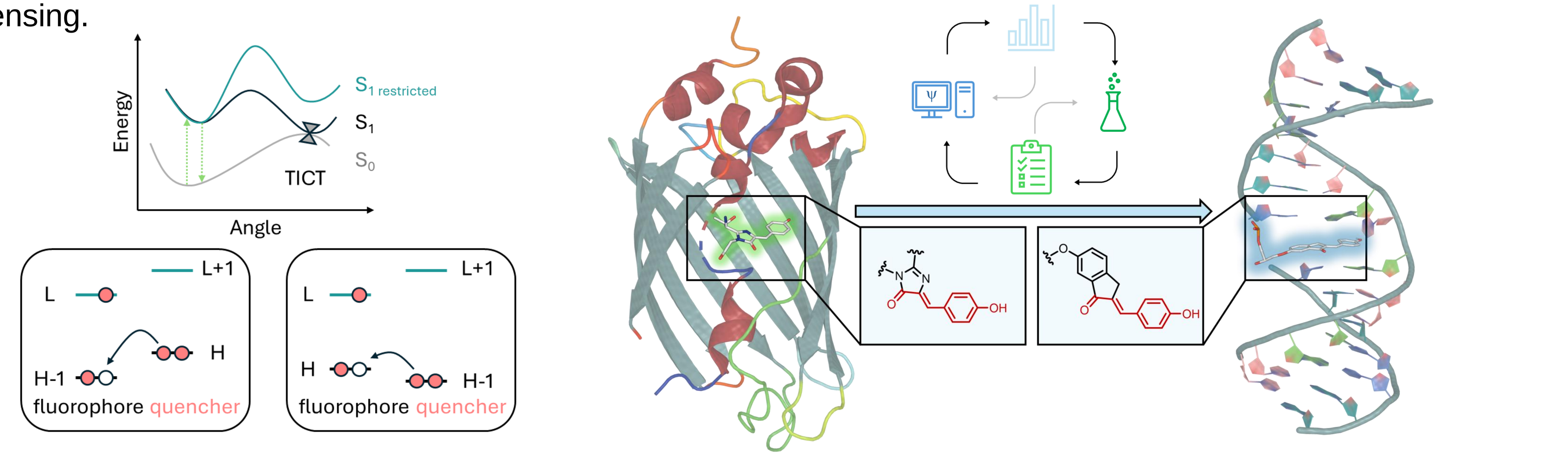
Introduction

Fluorescent proteins like GFP have transformed bioimaging by enabling real-time tracking of biomolecules.[1] While GFP-inspired fluorophores are well-established for proteins, nucleic acids (NAs) pose a greater challenge due to their inherent lack of fluorescence. There is a growing demand for emissive nucleotide analogs that integrate seamlessly into NAs while providing bright, environment-sensitive signals for biosensing. We present a modular synthesis strategy to directly install fluorescent molecular rotors (FMRs) into NA strands. Using 6-hydroxyindanone and aromatic aldehydes, we developed *O*-donor chalcone probes with large Stokes' shifts, dual excitation/emission wavelengths, and ratiometric capabilities. To guide probe design, we employed molecular dynamics (MD) simulations, (TD)-DFT calculations, and spectroscopy to reveal how structural and photophysical properties vary across DNA, DNA:RNA hybrids, and G-quadruplexes. These probes operate through twisted intramolecular charge transfer (TICT) and photoinduced electron transfer (PET) mechanisms. In flexible environments, TICT leads to quenched emission, but structural constraints, like duplex formation, restrict twisting and trigger strong fluorescence. PET modulation by NA topology or pH enables precise, light-up signaling. Compared to both traditional labels and previous GFP surrogates, these chalcone FMRs offer superior brightness, resolution, and tunability for next-generation NA biosensing.



Computational Methods

MD Simulations. The starting TBA15 GQ was extracted from the TBA-thrombin crystal structure (PDB: 4DII). The probe and noncanonical nucleic acid residues were parametrized using PyRED. MD simulations used the AMBER DNA (OL15), water (TIP4P-EW), and general amber (GAFF) force fields. The models were solvated in a truncated octahedral box water box with a 10 Å distance from solute. The system was neutralized with Na^+ and brought to a NaCl concentration of 150 mM. Four steps of minimization, six steps of heating, and five steps of equilibration was carried out prior to production. Constant molar, pressure, and temperature (NVT) ensemble MD simulations were run for 1000 ns with 3–5 replicates for each system using AMBER20 (DNA ends restrained). Data was extracted with the CPPTRAJ module of AMBER.

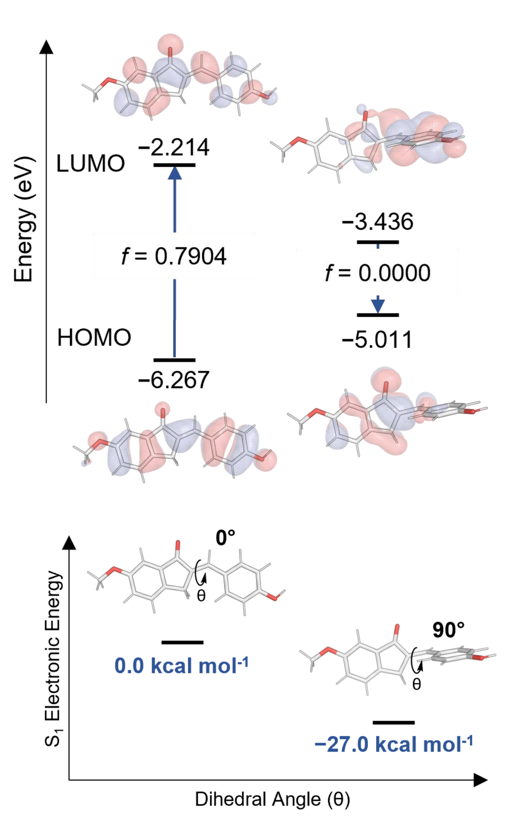
DFT on Free Probes. All calculations were performed with the Gaussian 16 (rev. C.01) suite of programs. The ground state geometries of were optimized using SMD-ωB97X-D3/def2-SVP (water). Single-point energy calculations were preformed using SMD-ωB97X-D3/def2-TZVP (water). Excited state geometries were optimized using TD-DFT/LR-SMD-ωB97X-D3/def2-SVPP (water). Excited state single-point energy calculations were preformed using TD-DFT/cLR-SMD/M06/def2-TZVP (water).

DFT on DNA–Probe Complexes. Hierarchical agglomerative clustering was carried out on the probe and flanking bases from the MD simulations. The clustered structures were optimized using SMD-ωB97X-D3/def2-SVP (water). Excited state single-point energy calculations were preformed using TD-DFT/cLR-SMD/ωB97X-D3/def2-TZVP (water). All calculations were performed with the Gaussian 16 (rev. C.01) suite of programs.

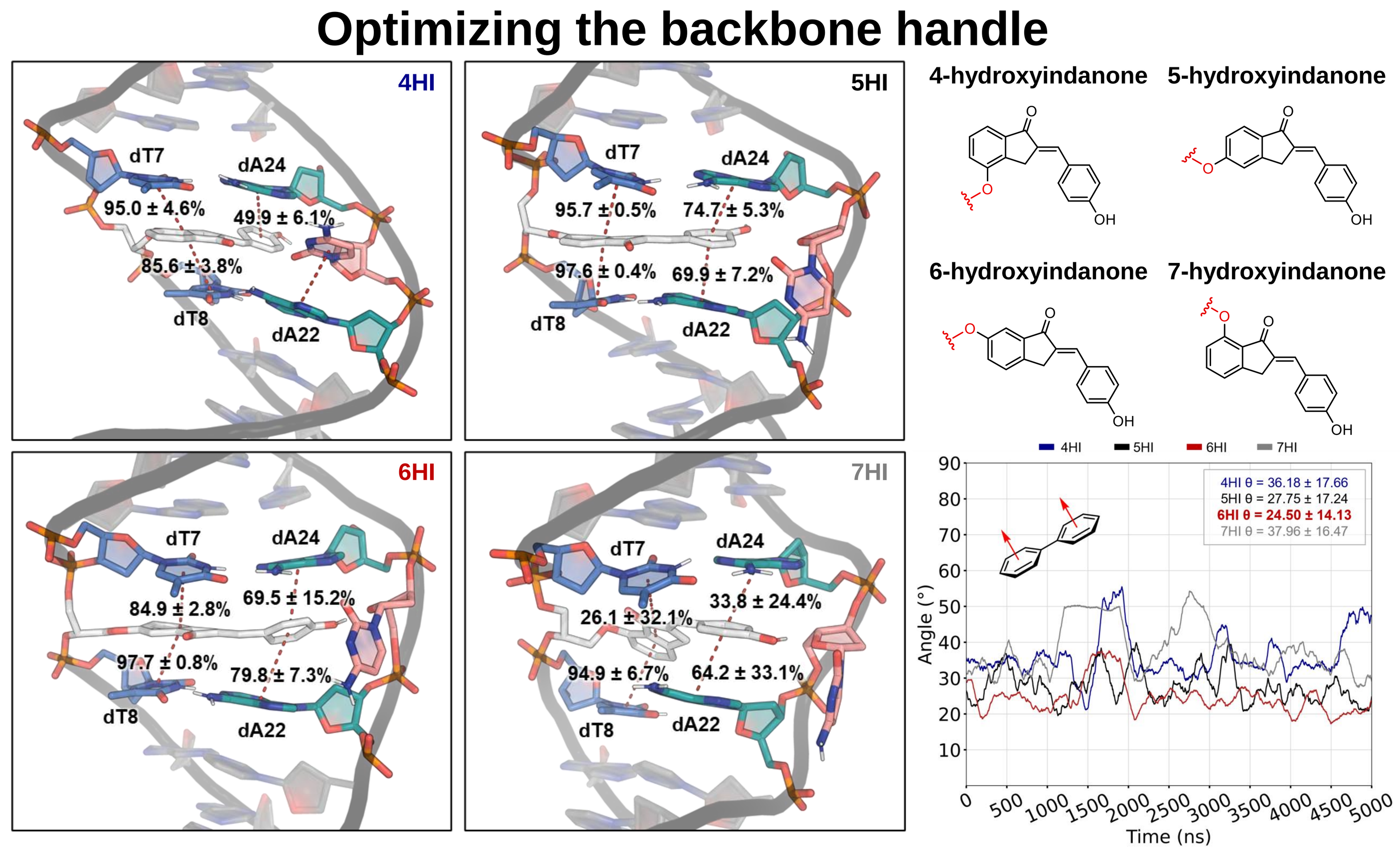
DFT on the TBA15 GQ–Probe. Hierarchical agglomerative clustering on the TBA15 GQ–probe from the MD simulations. The clustered structure was optimized using CPCMX-GFN2-xTB (water). Single-point energy calculations were preformed using CPCM-ωB97X-D3/def2-TZVP (water). All calculations were carried out with ORCA 6.0.0 and accelerated using the resolution of identity approximation (RI) for the Coulomb integrals, while the exchange terms were efficiently computed using the 'chain-of-spheres' (COSX) approximation with the def2/J Coulomb fitting auxiliary basis set.

Results

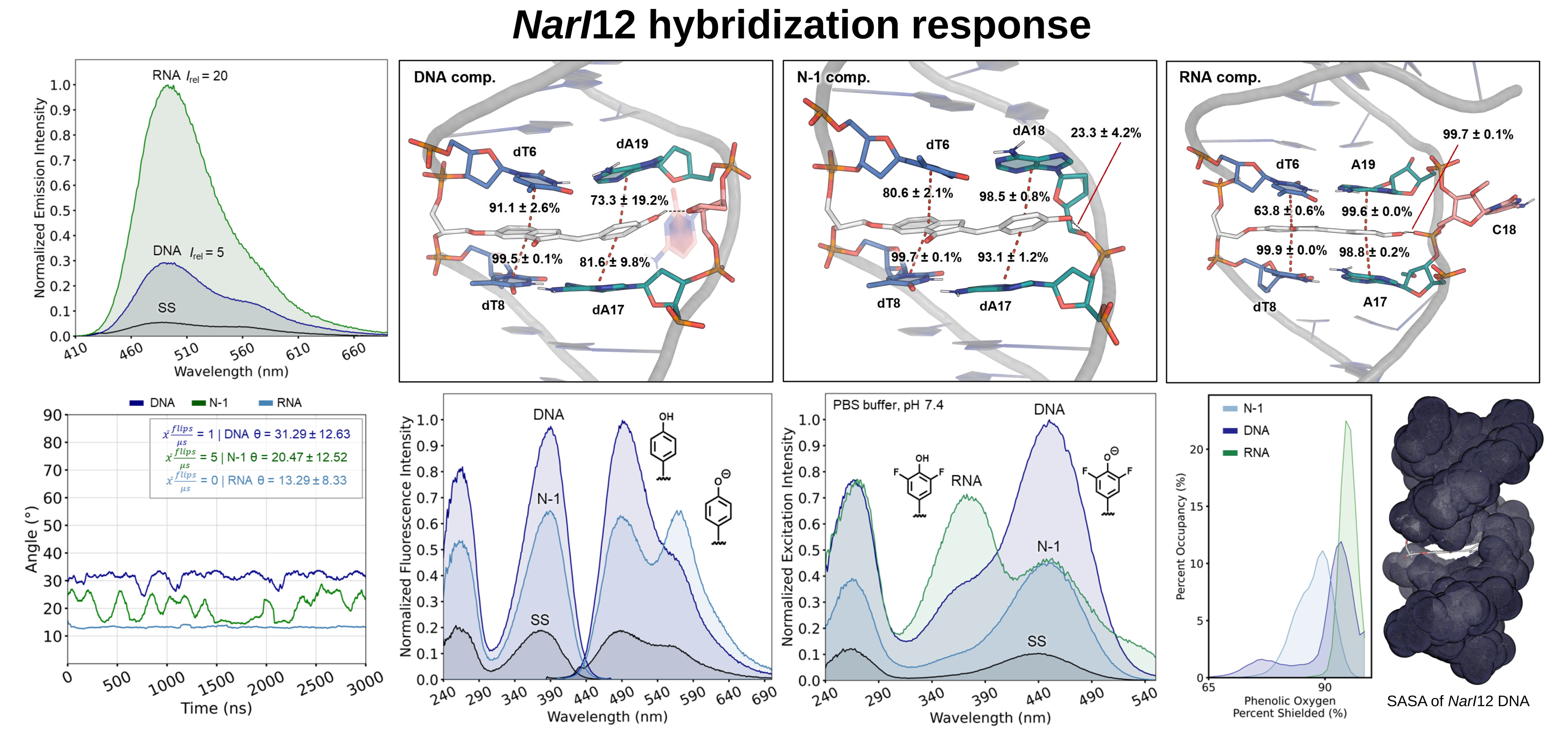
TICT formation



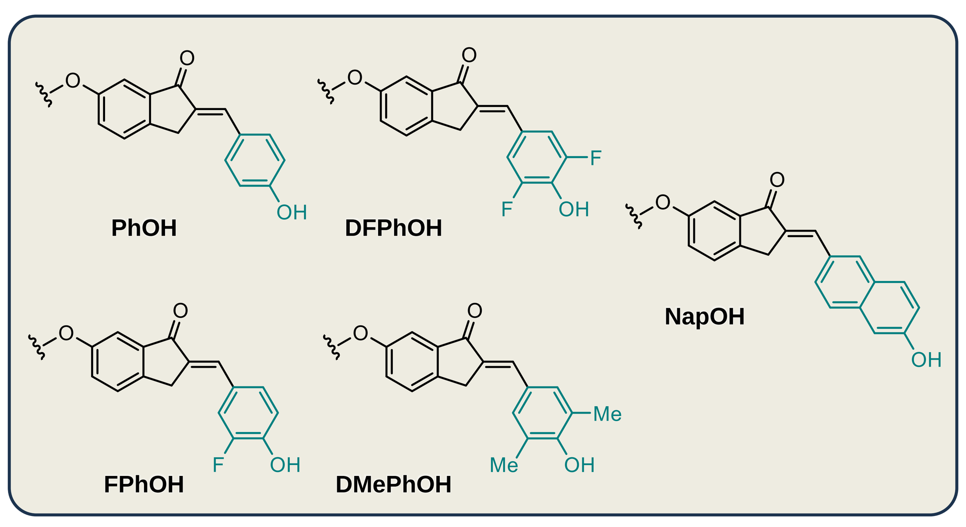
Optimizing the backbone handle



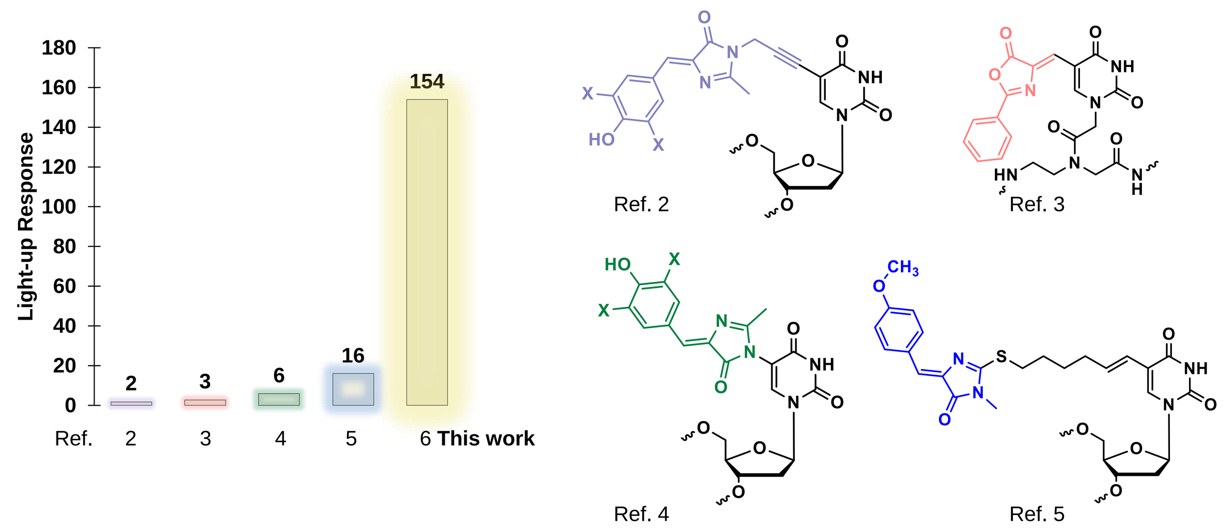
*NarI*12 hybridization response



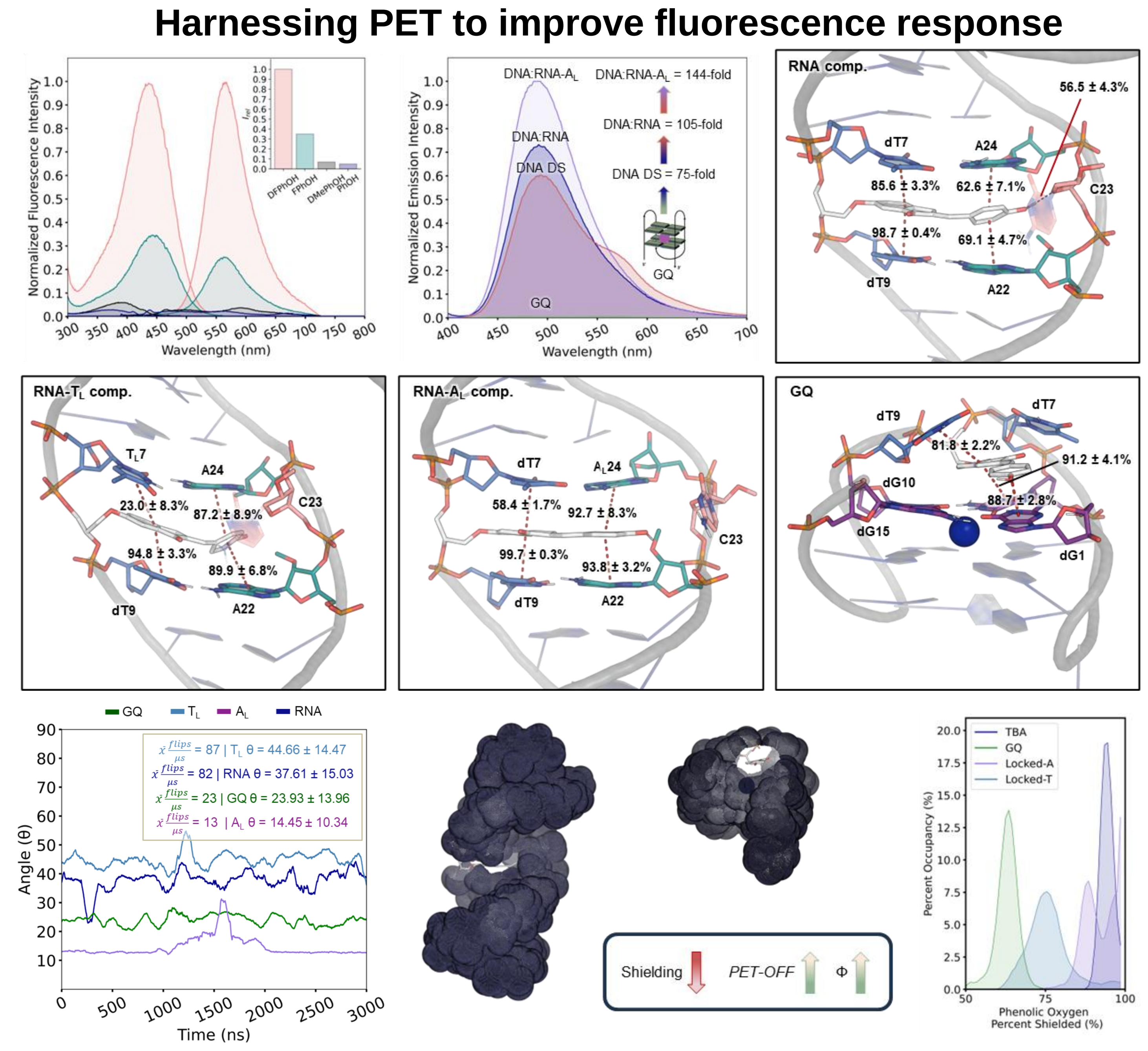
Fluorescent probe library



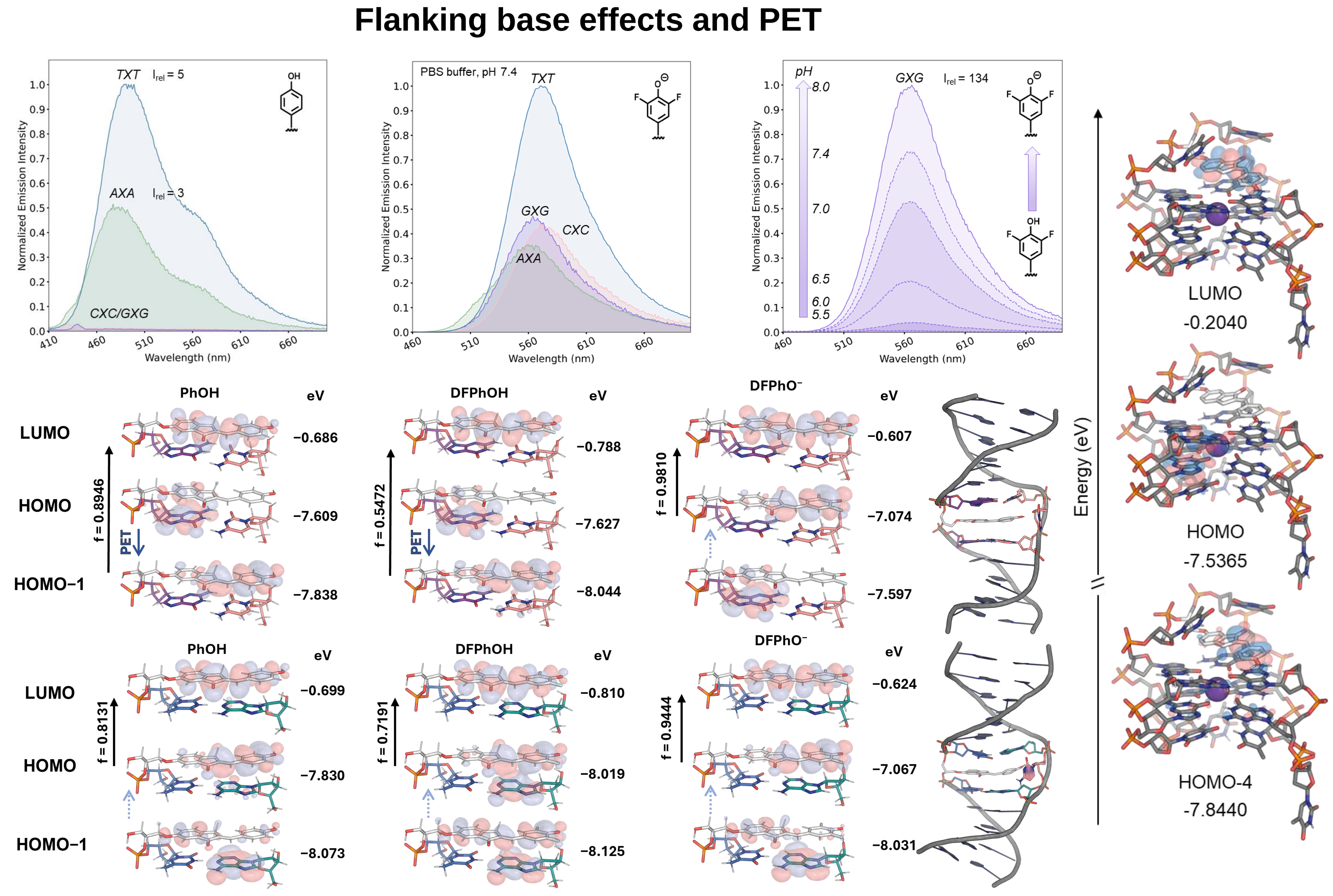
Comparison of nucleic acid GFP surrogates



Harnessing PET to improve fluorescence response



Flanking base effects and PET



Conclusions

Our synergistic approach led to the development of phenolic O-donor chalcone FMRs as tunable, high-performance GFP surrogates for nucleic acids. MD simulations identified 6-hydroxyindanone as the optimal handle, while (TD)-DFT calculations clarified the PET quenching observed with G-flanked probes. Harnessing this PET quenching process, large light-up responses through a PET ON-to-OFF switch were possible when transitioning between G-quadruplexes and duplexes, providing light-up responses up to 154-fold. TICT restriction upon hybridization drives strong fluorescence, and further in silico optimization of ribose linkers improved emission efficiency. Fluorinated variants introduced ratiometric excitation and pH-responsive behavior, with a 134-fold light up response within a duplex environment. Together, these findings establish these chalcone-based GFP FMRs as potentially powerful tools for biosensing and NA diagnostics. Moreover, the synergy between calculations and experiments not only enables precise prediction and validation of probe behavior across nucleic acid structures but also sets a powerful foundation for the rational design of next-generation fluorescent probes.

References

1. Sinkeldam, R. W.; Greco, N. J.; Tor, Y. *Chem. Rev.* **2010**, *110*, 2579.
2. Riedl, J.; Ménova, P.; Pohl, R.; Orság, P.; Fojta, M.; Hocek, M. *J. Org. Chem.* **2012**, *77*, 8287.
3. Chowdhury, M.; Turner, J. A.; Cappello, D.; Hajjami, M.; Hudson, R. H. E. *Org. Biomol. Chem.* **2023**, *21*, 9463.
4. Kanamori, T.; Takamura, A.; Tago, N.; Masaki, Y.; Ohkubo, A.; Sekine, M.; Seio, K. *Org. Biomol. Chem.* **2017**, *15*, 1190.
5. Saady, A.; Böttner, V.; Meng, M.; Varon, E.; Shav-Tal, Y.; Ducho, C.; Fischer, B. *Eur. J. Med. Chem.* **2019**, *173*, 99.
6. Regen, K. T.; Pounder, A.; Johnson R. E.; Murray, M. T.; Glowacki, H. X.; Wetmore, S. D.; Manderville, R. A. *Chem. Sci.* **2025**, *16*, 6468.

Our publication

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